

Fig. 7: An actual-size, single-side PCB for interfacing the Nokia colour LCD with the AVR microcontroller circuit

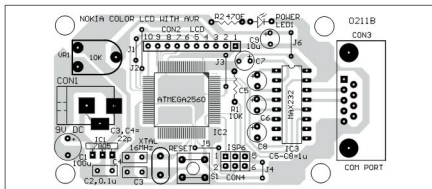


Fig. 8: Component layout for the PCB in Fig. 7

in your ATmega2560 application board.

Software program

The software for the microcontroller is written in 'C' language using the IAR Embedded Workbench integrated development environment. IAR Embedded Workbench is being developed by IAR Systems and ATMEL developers in parallel and hence it generates the optimised code which uses full 'C' coding capabilities of AVR devices. AVR development tools for embedded systems can be downloaded for free from IAR website www.iar.com. For details of IAR Embedded Workbench, you may refer to 'A Beginners' Guide to ATMEL AVR Development' article published in January issue of EFY.

To start the application, click 'IAR Embedded Workbench' icon in 'All Programs' menu of Windows. From the main menu, select Project→Create New Project and start an AVR Studio

compatible C project. Generate the Intel hex file for burning the code into Atmega2560 microcontroller as follows: Project→Options→Linker (under Category)→Extra Output→Generate Extra Output (to click)→Override Default (to click)→Type Project Name. hex→Output Format→intel-standard (to select).

Press 'ok' after all these settings are done. The source code for LCD interfacing (main.c) along with comments for instructions is included in this month's EFY-CD. Import the main.c file from the CD into the editor screen by clicking 'Add Files...' option in Project menu. Click 'Compile' option from Project menu to compile and generate main.hex code.

To send a command, a low bit is sent first before sending an 8-bit command, forming a 9-bit SPI command signal. Similarly, before sending an 8-bit data, a high bit is sent forming a 9-bit SPI data. You may refer to

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V regulator
- IC2 - ATmega2560 microcontroller
- IC3 - MAX232 RS-232 driver
- LED1 - 5mm light-emitting diode

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 10-kilo-ohm
- R2 - 470-ohm
- R3-R5 - 1-kilo-ohm
- R6-R8 - 2.2-kilo-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 100µF, 25V electrolytic
- C2 - 0.1µF ceramic
- C3, C4 - 22pF ceramic
- C5-C8 - 1µF, 16V electrolytic
- C9 - 10µF, 16V electrolytic

Miscellaneous:

- CON1 - DC connector
- CON2 - 10-pin berg strip female connector
- CON3 - 9-pin D-type female connector
- CON4 - ISP 6-pin male connector
- CON5 - 10-pin berg strip male connector
- CON6 - 10-pin connector for Nokia LCD
- S1 - Tactile switch
- S2 - Nokia 6100 colour LCD module

the datasheet of PCF8833 for various instructions used in the LCD driver chip. The code for this project (main.c) has been programmed as per these instructions.

If you can draw a point with a colour at any given location on the LCD screen, you can implement various graphics algorithms to draw a circle, line, filled rectangle, etc. You can have the AVR microcontroller fitted with a 16.00MHz external crystal oscillator, a serial RS-232 driver connecting a serial port and a reset arrangement to run the code.

Downloading the code into Atmega2560. To program the Atmega2560, connect the 6-wire cable between ISP connectors of the STK500 board (marked as ISP6) and the target board (marked as ISP con). Connect a serial cable from 'RS232 CTRL' connector on the STK500 board to a COM port on the PC. Now start AVR Studio 4.0 without opening any project file.

Optionally, you may proceed as follows: Main Menu→Tools→Program